The node roles are as follows:

- *Master node*—Involved in high-level operations such as creating and deleting indexes, node operations, and other admin-related jobs for cluster management. These admin operations are lightweight processes; hence, one master is enough for an entire cluster. If this master node crashes, the cluster elects one of the other nodes as the master, so the baton passes. Master nodes don't participate in document CRUD operations, but the master node knows the location of the documents.
- *Data node*—Where indexing, searching, deleting, and other document-related operations happen. These nodes host indexed documents. Once an index request is received, the data nodes jump into action to save the document to its index by calling a writer on the Lucene segment. As you can imagine, they talk to the disk frequently during CRUD operations; hence, they are disk I/O- and memory-intensive operations.

NOTE Specific variants of the data node role are used when we deploy multi-tiered deployments: the `data_hot`, `data_cold`, `data_warm`, and `data_frozen` roles. We go over them in chapter 14.

- *Ingest node*—Handles ingest operations such as transformations and enrichment before indexing kicks in. Documents ingested via a pipeline operation (for example, processing Word or PDF documents) can be put through additional processing before being indexed.
- *Machine learning node*—As the name indicates, executes machine learning algorithms and detects anomalies. It is part of a commercial license, so you must purchase an X-Pack license to enable machine learning capabilities.
- *Transform node*—The latest addition to the list. A transform node is used for aggregated summaries of data. It is required for carrying out transform API invocations to create (transform) new indexes that are pivoted based on existing indexes.
- *Coordinating node*—While roles are assigned to a node by the user on purpose (or by default), there's one special role that all nodes take on, regardless of user intervention: coordinating node (or coordinator). As the name suggests, the coordinator looks after client requests end to end. When a request is made of Elasticsearch, one of these nodes picks up the request and dons the coordinator's hat. After accepting the requests, the coordinator asks the other nodes in the cluster to process the request. It awaits the response before collecting and collating the results and sending them back to the client. It essentially acts as a work manager, distributing incoming requests to appropriate nodes and responding to the client.

CONFIGURING ROLES

When we start up Elasticsearch in development mode, the node is by default set with `master`, `data`, and `ingest` roles (and each node is by default a coordinator—there is no special flag to enable or disable a coordinating node). We can configure these